CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (IT/CE) No Candidate

Dec 2013

IT310 Artificial Intelligence (A.I.) (Elective-I)

Date: 07.12.2013, Saturday   Time: 01:30 p.m. To 04:30 p.m.   Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

a) What do you mean by search space and problem space? How to search a problem (state) space? 3

b) What is admissibility in a heuristic search algorithm? What is an admissible heuristic for A* algorithm? 3

c) Differentiate between Breadth First Search (BFS) and Depth First Search (DFS). Which is better when the memory is limited? Give reason. 3

d) Explain the various blocks in PROLOG program. Which out of them are compulsory and which are optional? 2

Q-2

[A] Determine whether the following sentences are tautology, contradiction or neither. 4

i) $(P \wedge (P \rightarrow Q)) \rightarrow Q$ ii) $(P \wedge Q) \wedge \neg(P \wedge Q)$

[B] Suggest a good heuristic function for the following problems: 4

i) TSP   ii) Tic Tac Toe

[C] Will the following literals Unify? If yes, then write the most general unifier. 4

W={ P(a, x , f(g(y))), P (z, f(z),f(u))}

OR

Q-2

[A] Briefly describe Bayesian belief nets and how are they used to classify items for a given problem. 4

[B] What are the distinctive characteristics of multi layer perceptron? 4

[C] Justify- NLP is considered hallmark of human intelligence. 4

Q-3

[A] Write note on-issues in knowledge representation. 4

[B] Write down the steps involved in the knowledge acquisition process of an Expert System. 4

[C] Give a CFG for ENGLISH language and give leftmost derivation for the sentence: 4

'That girl plucked the flower'.

OR

Q-3

[A] Describe briefly the frame based knowledge representation by explaining its advantages and disadvantages. 4

[B] Write a prolog program to read N elements and store into the list. 4

[C] What is the significance of planning in AI systems? Explain the main components of a planning system.. 4

SECTION-II

Q-4

a) Production system and control strategies are two approaches of problem solving. Discuss the important characteristics of production system and control strategy. 3

b) Will the following literals Unify? If yes, then write the most general unifier. 4

1. hate(x,y)
2. hate(Marcus,z)

c) Compare Procedural Vs Declarative knowledge. 4

Q-5

[A] Explain the Expert System development process. 4

[B] What are the potentialities and limitations of Neural Networks? 4

[C] What do you mean by grammar and parsing? Why natural language understanding is difficult problem? 4

OR

Q-5

[A] Explain Cut, Recursion and Negation in brief. Give suitable example from PROLOG. 4

[B] Solve the water jugs problem. Given two jugs of 4 l and 3 l respectively, fill the 4 l jug with 2 l of water. Find a good heuristic and perform hill climbing. 4

[C] Draw the conceptual Dependency (CD) representation for: *John punched Bill*. 4

Q-6

[A] Convert these sentences to predicate logic. Using the logical rules, proof by resolution that prove "***whether John eats Kumquats or not***". 4

(a) John likes fruit.
(b) All fruits are not liked by John.
(c) Kumquats are fruits.
(d) John eats only those fruits, which he likes.

[B] Write a prolog program for generating the sublists of the given list. 4

OR

[B] Trace the execution of the constraint satisfaction procedure in solving the crypt arithmetic problem: 4

DONALD
\+
GERALD
———
ROBERT

[C] Write a prolog program to add all elements of a given integer list. 4

OR

[C] What is Back-propagation error? How is it used for batch training and incremental training? What learning rate should be used for back propagation? 4